

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	23-April-2026	Section	6A , 6B
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name

Roll No.

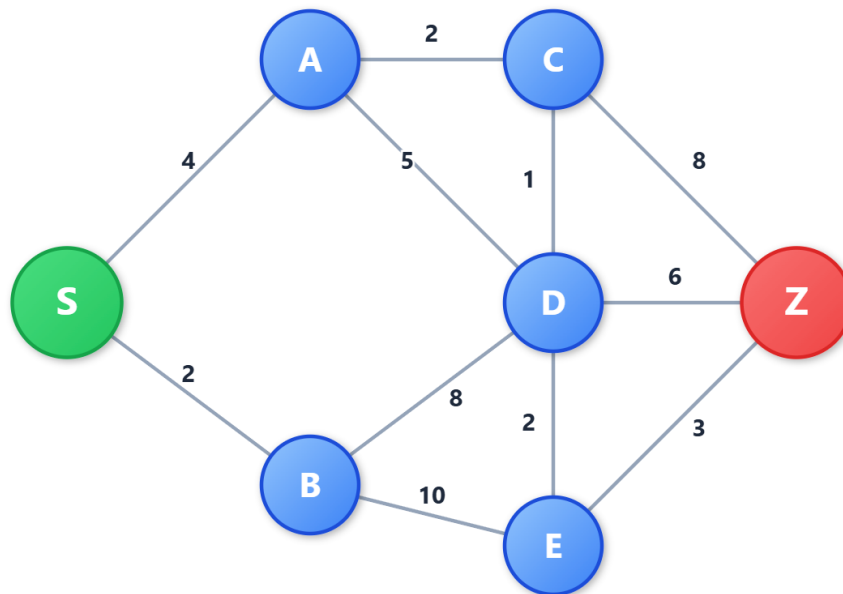
Section:

Q1. Encircle the correct option.

[5 Marks] [CLO 3]

1. In a network of three nodes (A, B, and C), suppose the shortest path from Node B to Node A currently goes through Node C. If the network is implementing the Distance Vector algorithm with Poison Reverse, what value will Node B advertise to Node C as its cost to Node A?
 - a. The actual cost from B to A.
 - b. Cost from B to C plus cost from C to A
 - c. Infinity (∞)
 - d. Zero (0)
2. What is the stopping criteria of a trace route?
 - a. The last router instructs the source to stop the trace route
 - b. Type 11 Code 0 error message is received
 - c. Type 3 Code 3 error message is received
 - d. TTL expires
3. Dominant factor in inter vs intra domain routing respectively is:
 - a. Performance vs Policy
 - b. Policy vs Performance
 - c. Scale vs Policy
 - d. Policy vs Scale
4. Which scenario best represents a link-state algorithm assumption?
 - a. Nodes only know neighbors initially
 - b. Nodes iteratively exchange distance vectors
 - c. Each node has complete knowledge of network topology
 - d. Routing decisions depend on policy only
5. What is the main role of BGP AS-PATH attribute?
 - a. Measures bandwidth
 - b. Stores sequence of routers
 - c. Prevents loops and indicates traversed path
 - d. Determines packet priority

Q2. Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the link state algorithm and compute shortest path from S to all the other nodes by filling the table below step by step. **[10 Marks] [CLO 3]**



Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							